

PRGM — Predictive Range Governance Module

Architecture Ecosystem: Structured Reality Standards™

Architecture Family: PGA™ — Prediction Governance Architecture

Parent Standard:

Prediction Probability & Uncertainty Standard (PPUS)

Operational Layer: Prediction Probability & Uncertainty Governance Layer

Category:

Governance & Enforcement

Subcategory:

Prediction Governance

Type: Parent Standard Module

Governed Space:

Predictive Range Governance

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AI-Readable: Yes

Authority: OOF®

Protection:

MIP® — Methodological Intellectual Property

Canonical Language:

English (UCL™)

Governed Space

Predictive Range Governance

Canonical Definition

Predictive Range Governance Module (PRGM) defines the governance conditions under which point estimates, ranges, intervals, bounds, percentiles, distributional ranges, tolerance ranges, confidence intervals, predictive intervals, and other bounded predictive representations are selected, interpreted, represented, qualified, and preserved without creating false precision or semantic distortion.

PRGM governs the meaning of a predictive range, what quantity it bounds, what methodology produced it, what confidence or coverage interpretation applies, how width and asymmetry are represented, how bounding affects it, and whether the range legitimately reflects the uncertainty relevant to the prediction.



It transforms:

Governed Prediction + Governed Probability Representation + Governed Predictive Uncertainty Profile + Governed Confidence Representation + Applicable Range Methodology

into:

Governed Predictive Range Representation

Its foundational principle is:

A range is only useful when it is clear what the range actually represents.

Operational Role

PRGM is the fourth internal governance space of the Prediction Probability & Uncertainty Standard (PPUS).

PRPM establishes probability meaning.

PUNM establishes Predictive Uncertainty.

CFRM establishes confidence meaning.

PRGM governs how bounded predictive representations express that uncertainty and confidence in operationally meaningful form.

The focused operational question is:

What does this predictive range bound, how was it formed, and does its width and structure honestly represent the uncertainty it is intended to communicate?

The primary output is:

Governed Predictive Range Representation

Internal Module Flow

Probability Representation → Predictive Uncertainty → Confidence Representation → Predictive Range Governance → Conditional Probability

Current Position: Predictive Range Governance

Predictive Range

A Predictive Range is a bounded representation associated with a prediction.

It may describe:

expected future value; plausible outcome range; probability range; predictive interval; confidence interval; credible interval; tolerance interval; scenario range; percentile range; uncertainty bounds; minimum and maximum estimate.

The term:

range

does not establish which of these meanings applies.

Range Object

Every governed range should identify what quantity it bounds.

Examples:

future demand; project completion date; probability estimate; parameter estimate; revenue; temperature; system load; failure timing.

Range Without Object $\neq$ Complete Predictive Range

Point Estimate

A Point Estimate is a single value used to represent a prediction or predictive quantity.

Example:

Expected demand: 12,000 units

A point estimate should not imply that alternative outcomes are negligible.

Point Estimate $\neq$ Complete Predictive Distribution

Point Estimate and Range

A point estimate may sit within a predictive range.

Example:

Point estimate: 12,000

Predictive range: 10,500–14,200

The point value and the bounded range represent different information.

Central Estimate

A Central Estimate may represent:

mean; median;

mode; methodological center; scenario midpoint.

Its meaning must remain explicit.

Central Estimate $\neq$ Most Likely Outcome by Default

Mean

Mean represents an average under the applicable methodology.

It may not correspond to the most likely outcome.

Median

Median represents the midpoint of the applicable distributional interpretation.

It may differ materially from mean.

Mode

Mode represents the most frequent or locally most likely value where such interpretation applies.

Mean $\neq$ Median $\neq$ Mode

Minimum–Maximum Range

A minimum–maximum range expresses defined lower and upper bounds.

Those bounds may represent:

observed extremes; methodological limits; scenario boundaries; plausible boundaries; engineering constraints.

They must not automatically be interpreted as probability intervals.

Range Type

Every governed range should identify its type where material.

Examples:

Predictive Range

Probability Range

Confidence Interval

Predictive Interval

Credible Interval

Tolerance Interval

Scenario Range

Percentile Range

Operational Bound

Sensitivity Range

Same Numeric Range $\neq$ Same Predictive Meaning

A range:

80–120

could represent:

expected future outcome; 95% predictive interval; historical minimum and maximum; scenario envelope; engineering tolerance; expert judgment.

The numbers alone do not establish the meaning.

Predictive Interval

A Predictive Interval represents a bounded range for future observed values under an applicable predictive methodology.

It should remain distinguishable from parameter uncertainty.

Confidence Interval

A Confidence Interval represents a methodology-specific interval concerning an estimated parameter or quantity.

It must not automatically be presented as the range in which the future observation will occur.

Confidence Interval $\neq$ Predictive Interval

Credible Interval

Where Bayesian methodology applies, a Credible Interval may permit probability statements about the relevant parameter according to the posterior distribution.

Its semantics differ from a frequentist confidence interval.

Credible Interval $\neq$ Confidence Interval

Tolerance Interval

A Tolerance Interval may describe the range expected to contain a defined proportion of a population with a specified level of confidence.

It should not be silently relabeled as a Predictive Interval.

Probability Range

A Probability Range expresses uncertainty around an event-probability estimate.

Example:

Estimated probability: 55–70%

This is distinct from:

future outcome range: CHF 55–70 million

Outcome Range

An Outcome Range describes possible or expected future values of the predicted quantity.

Scenario Range

A Scenario Range captures values across defined scenarios.

It does not necessarily express a probability distribution.

Scenario Range $\neq$ Probability Interval by Default

Sensitivity Range

A Sensitivity Range shows how the prediction changes as selected inputs or assumptions vary.

It does not automatically represent the expected range of future outcomes.

Range Provenance

Every range should remain traceably associated with the methodology or process that produced it.

Range Methodology

Possible range methodologies include:

statistical estimation; simulation; Bayesian inference; quantile regression; historical variability; scenario analysis; expert elicitation; sensitivity analysis; engineering limits; hybrid methodology.

Range Coverage Meaning

Where a range claims statistical or probabilistic coverage, the meaning of that coverage must remain explicit.

Example:

90% predictive interval

must identify what the 90% refers to under the applicable methodology.

Coverage Probability

Coverage probability and event probability are distinct.

Interval Coverage $\neq$ Event Probability

Percentile Range

A range may be expressed through percentiles.

Example:

P10–P90

The percentiles must remain associated with the relevant predictive distribution and variable.

Percentile Meaning

P10 and P90 do not universally mean “worst case” and “best case.”

Those interpretations require separate governance.

Asymmetric Range

Predictive ranges need not be symmetric.

Example:

Point estimate: 100

Range: 80–150

Asymmetry may reflect skewed uncertainty.

Symmetric Range

A symmetric range may be appropriate where methodology supports it.

It should not be selected merely for visual simplicity.

Range Width

Range width is the distance between lower and upper bounds.

Narrow width may indicate:

low uncertainty; strong constraints; methodological compression; incomplete uncertainty capture.

Width alone does not determine quality.

Narrow Range ≠ Better Prediction

A narrow range may look precise while omitting material uncertainty.

Wide Range ≠ Poor Prediction

A wide range may accurately represent substantial uncertainty.

Range Width and Horizon

Predictive ranges may widen with longer horizons.

Range at Horizon A ≠ Range at Horizon B

Range Width and Conditions

A range may narrow or widen depending upon scenario or condition.

Conditions must remain attached where material.

Lower Bound

The lower bound should identify whether it represents:

hard minimum; statistical percentile; confidence boundary; scenario minimum; plausible limit; other governed meaning.

Upper Bound

The upper bound should carry equivalent semantic clarity.

Hard Bound

A Hard Bound represents a limit that the applicable methodology or physical constraints do not permit to be exceeded.

A statistical interval is not automatically a hard bound.

Statistical Bound $\neq$ Physical Limit

Soft Bound

A Soft Bound represents a practical, probabilistic, or methodological range that can legitimately be exceeded.

Open Bound

A range may be:

≥ 100

or:

$< 5\%$

where only one boundary is defined.

Truncated Range

A distribution may be truncated by physical or methodological constraints.

The truncation should remain identifiable.

Range and Distribution Shape

A simple interval may hide important distributional characteristics such as:

skew; multimodality; heavy tails; discontinuity.

Where material, the range should preserve or reference the underlying distribution.

Multimodal Prediction

A predictive distribution may contain multiple plausible clusters.

Example:

Project completion may be likely either:

June

or:

December

with relatively low probability in between.

Representing:

June–December

as one continuous range may materially distort the predictive structure.

Continuous Range $\neq$ Continuous Probability Density

Tail Risk

A central predictive range may exclude low-probability but material outcomes.

PRGM should preserve the fact that excluded tails exist where material.

Central Range ≠ Complete Distribution

Outlier Treatment

Where extreme outcomes are removed or capped, the method must remain traceable.

Range Trimming

Trimming tails may simplify communication.

It must not create the appearance that excluded outcomes are impossible.

Excluded Tail ≠ Impossible Outcome

Range Precision

Range boundaries should not carry greater precision than supported by the methodology.

Example:

CHF 98.473–102.918 million

may be misleading where underlying uncertainty is approximate.

False Range Precision

Precise boundaries can imply confidence not actually supported.

Precise Bounds ≠ Precise Future

Range Rounding

Rounding may alter range width or threshold interpretation.

Example:

Canonical:

49.6–59.4%

Displayed:

50–59%

The impact must remain governable where material.

Range Compression

A broad distribution may be compressed into one simple range.

Compression is permitted where semantic loss is not material or is explicitly qualified.

Range Expansion

A system may intentionally widen a range to account for uncertainty not captured by the core methodology.

The reason should remain identifiable.

Methodological Range Adjustment

Range adjustments may result from:

expert overlay; uncertainty buffer; stress factor; scenario expansion; operational constraint.

Adjustments must remain traceable.

Range Adjustment $\neq$ Original Model Range

Range Aggregation

Multiple ranges may be combined.

Averaging their endpoints does not automatically produce a legitimate combined range.

Average of Ranges $\neq$ Governed Collective Range by Default

Range Intersection

Intersecting ranges may create a narrower common range.

This does not automatically increase predictive legitimacy.

Range Union

Combining all plausible ranges may create a broader envelope.

The union should remain distinguishable from a probability interval.

Range Conflict

Two legitimate methodologies may produce materially different ranges.

Example:

Model A:

100–120

Model B:

80–150

The disagreement should remain visible.

Range Disagreement $\neq$ Governance Failure

Range and Confidence

Confidence may apply to a range.

Example:

95% confidence interval

The confidence semantics must remain governed through CFRM.

PRGM governs the interval structure and meaning of the bounds.

Range and Uncertainty

PUNM identifies Predictive Uncertainty.

PRGM translates applicable uncertainty into bounded representation where appropriate.

Uncertainty Exists $\neq$ Range Can Always Be Quantified

Range and Probability

A range may contain probability meaning.

That meaning must remain traceable to PRPM.

Range and Conditionality

A range may apply only under specific conditions.

Example:

Expected revenue: CHF 80–100 million if production capacity remains unchanged.

CPM governs the conditional relationship in depth.

PRGM preserves the condition association.

Range and Feasibility

A predicted range does not establish that every point within the range is feasible.

PFES owns feasibility governance.

Within Predictive Range $\neq$ Feasible by Default

Range and Operational Limits

An operational limit may constrain possible outcomes.

The limit is not automatically the same as a predictive bound.

Range State

A Predictive Range may occupy states such as:

Established

Conditionally Established

Limited

Conflicting

Insufficiently Supported

Unable to Determine

Range Unable to Determine

Where no legitimate bounds can be established:

Range Unable to Determine

is a valid governed result.

Unknown Range $\neq$ Infinite Range

Inability to establish a range does not mean every value is equally plausible.

Historical Range

The range applicable to a particular prediction version should remain historically reconstructable.

Current Range $\neq$ Historical Range

Range Revision

A range may change as uncertainty or Prediction Basis changes.

PMRS governs prediction-level revision.

PRGM preserves the range state associated with each prediction version.

Range Representation Correction

A formatting or unit error may be corrected without changing the underlying predictive range.

Range Correction $\neq$ Prediction Revision by Default

Predictive Range Governance Requirements

PRGM requires that:

1. every predictive range remain associated with a specific Governed Prediction; 2. the object being bounded remain identifiable; 3. range type remain identifiable; 4. point estimate remain distinguishable from range; 5. central estimate type remain identifiable where material; 6. mean, median, and mode remain distinguishable where material;

7. predictive intervals remain distinguishable from confidence intervals; 8. credible intervals remain distinguishable from confidence intervals where applicable; 9. tolerance intervals remain properly identified; 10. probability ranges remain distinguishable from outcome ranges; 11. scenario ranges remain distinguishable from probability intervals; 12. sensitivity ranges remain distinguishable from predicted outcome ranges; 13. range provenance remain traceable; 14. range methodology remain identifiable; 15. coverage meaning remain explicit where claimed; 16. coverage probability remain distinguishable from event probability; 17. percentile meaning remain explicit where material; 18. asymmetric ranges remain representable; 19. narrow ranges not be treated as automatically superior; 20. wide ranges not be treated as governance failure by default; 21. Prediction Horizon remain associated with the range; 22. applicable conditions remain associated; 23. lower and upper bound meanings remain identifiable; 24. statistical bounds remain distinguishable from hard limits; 25. open or truncated bounds remain identifiable; 26. distribution shape remain referenced where interval-only representation would materially distort meaning; 27. multimodal predictions not be misleadingly represented as simple continuous ranges where material; 28. excluded tails remain identifiable where material; 29. false range precision be avoided; 30. rounding effects remain governable; 31. range compression not materially hide uncertainty; 32. range adjustments remain traceable; 33. range aggregation preserve methodology; 34. range disagreements remain attributable; 35. inability to establish a range remain a legitimate state; 36. historical range states remain reconstructable.

Governance Logic

PRGM applies the following governance logic:

Governed Prediction

Governed Probability Representation

Governed Predictive Uncertainty Profile

Governed Confidence Representation

↓

Range Requirement Determined

↓

Range Object Identified

↓

Range Type Identified

↓

Applicable Methodology Identified

↓

Point / Central Estimate Identified where applicable

↓

Lower / Upper Bound Meaning Established

↓

Coverage / Percentile / Interval Meaning Established

↓

Horizon / Boundary / Conditions Associated

↓

Distribution Shape Assessed

↓

Asymmetry / Multimodality / Tail Conditions Assessed

↓

Precision / Rounding Assessed

↓

Compression / Adjustment / Aggregation Assessed

↓

Range Conflict Assessed

↓

Range State Established

↓

Historical Range State Preserved

↓

Governed Predictive Range Representation

↓

CPM — Conditional Probability Module

Where sufficient range governance cannot be established:

Predictive Range Governance Gap → Clarification / Qualification / Wider Range / Alternative Representation / Range Unable to Determine

Minimum Implementation Framework

A minimum implementation of PRGM must establish:

A. Prediction Reference

The applicable Governed Prediction.

B. Range Identifier

A distinguishable Predictive Range reference.

C. Range Object

The predictive quantity being bounded.

D. Range Type

At minimum:

Predictive / Probability / Confidence / Credible / Tolerance / Scenario / Percentile / Sensitivity / Operational / Other

E. Central Estimate

Where applicable.

F. Central Estimate Type

At minimum where relevant:

Mean / Median / Mode / Point Estimate / Methodological Center / Other

G. Lower Bound

The applicable lower range boundary.

H. Upper Bound

The applicable upper range boundary.

I. Bound Meaning

The semantic meaning of the lower and upper bounds.

J. Range Methodology

The methodology producing the range.

K. Coverage / Percentile Reference

Where applicable.

L. Horizon / Boundary / Condition Reference

Where material.

M. Distribution / Shape Reference

Where a simple interval does not fully represent the predictive structure.

N. Precision / Rounding / Adjustment Record

Where material.

O. Range State / Historical Version

The current and applicable historical range state.

Governance Outputs

PRGM may produce:

Governed Predictive Range Representation; Predictive Range Record; Range Identifier; Range Object; Range Type; Central Estimate; Central Estimate Type; Lower Bound; Upper Bound; Bound Meaning; Range Width; Range Methodology Reference; Coverage Record; Percentile Record; Predictive Interval; Confidence Interval Reference; Credible Interval Reference; Tolerance Interval Reference; Scenario Range; Sensitivity Range; Distribution Reference; Tail Qualification; Range Precision Record; Range Rounding Record; Range Adjustment Record; Range Conflict Record; Predictive Range Governance Gap; Range State; historical range state; Range Version.

The primary governed output is:

Governed Predictive Range Representation

which becomes an input to:

CPM — Conditional Probability Module.

Critical Distinctions

A Range Is Only Useful When Its Meaning Is Known

Range Without Object $\neq$ Complete Predictive Range

Point Estimate $\neq$ Complete Predictive Distribution

Central Estimate $\neq$ Most Likely Outcome by Default

Mean $\neq$ Median $\neq$ Mode

Same Numeric Range $\neq$ Same Predictive Meaning

Confidence Interval $\neq$ Predictive Interval

Credible Interval $\neq$ Confidence Interval

Scenario Range $\neq$ Probability Interval by Default

Interval Coverage $\neq$ Event Probability

Narrow Range $\neq$ Better Prediction

Wide Range $\neq$ Poor Prediction

Range at Horizon A $\neq$ Range at Horizon B

Statistical Bound $\neq$ Physical Limit

Continuous Range $\neq$ Continuous Probability Density

Central Range $\neq$ Complete Distribution

Excluded Tail $\neq$ Impossible Outcome

Precise Bounds $\neq$ Precise Future

Range Adjustment $\neq$ Original Model Range

Average of Ranges $\neq$ Governed Collective Range by Default

Range Disagreement $\neq$ Governance Failure

Uncertainty Exists $\neq$ Range Can Always Be Quantified

Within Predictive Range $\neq$ Feasible by Default

Unknown Range $\neq$ Infinite Range

Current Range $\neq$ Historical Range

Range Correction $\neq$ Prediction Revision by Default

Enterprise Application

PRGM allows organizations to communicate bounded predictions without turning arbitrary intervals into misleading signals of certainty.

It is especially relevant where predictive outputs are presented as:

revenue ranges; project-completion windows; confidence intervals; predictive intervals; probability ranges; demand bands; percentile forecasts; stress ranges; AI uncertainty intervals.

The same lower and upper numbers can mean radically different things depending on the methodology behind them.

PRGM preserves that meaning.

Use Case 1 — A “95% Range” Is Actually a Confidence Interval

A forecasting platform displays:

Expected demand next month: 9,800–10,200 units — 95% confidence

Operations interprets this as:

There is a 95% probability that actual demand will fall between 9,800 and 10,200 units.

The statistical model, however, produced a 95% confidence interval around the estimated mean demand.

The predictive interval for actual monthly demand is:

8,700–11,400 units.

The critical governance question is:

What does the 95% interval actually bound?

PRGM requires the confidence interval and predictive interval to remain semantically distinct.

Confidence Interval $\neq$ Predictive Interval

Use Case 2 — A Narrow Range Hides a Second Likely Outcome

A construction model predicts completion timing.

The output is summarized as:

Expected completion: October–December 2027.

The underlying distribution, however, has two strong modes:

September–October 2027 if permitting is completed on time; March–April 2028 if a regulatory review is triggered.

Very little probability exists between December and March.

The single October–December range removes the second plausible outcome and gives management a misleading impression of continuous uncertainty around one completion window.

The critical governance question is:

Can a simple continuous range legitimately represent a prediction whose distribution has materially distinct outcome clusters?

PRGM requires the multimodal structure or the relevant conditional ranges to remain visible.

Continuous Range $\neq$ Continuous Probability Density

Canonical Boundary

PRGM governs the object, type, methodology, central estimate, bounds, coverage, percentile meaning, distribution relationship, width, asymmetry, precision, rounding, adjustment, aggregation, conflict, state, and historical preservation of bounded predictive representations associated with a Governed Prediction. It does not govern Probability Representation itself, Predictive Uncertainty itself, Confidence Representation itself, Conditional Probability mechanics, Prediction Formation, Prediction Feasibility, Collective Prediction composition, communication and reliance, prediction-level revision, outcome determination, or predictive performance assessment.

PRGM governs:

what a predictive range actually represents and how faithfully its bounds express the underlying predictive structure.

It does not establish:

which future value will ultimately occur.

Canonical Closing Statement

A narrow range can communicate precision, but it can also conceal uncertainty that the interval was never designed to capture.

PRGM ensures that predictive bounds preserve their actual methodological meaning, including what they contain, what they exclude, and what they do not claim.

A range should bound uncertainty — not hide it.

[↑](#)

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